

RESEARCH

Relationship between maternal perceptions and eating behaviors in relation to the nutritional status of schoolchildren**Relación de la percepción materna y conductas alimentarias sobre el estado nutricional del escolar****Relação entre a percepção materna e os comportamentos alimentares sobre o estado nutricional dos escolares**Luz Elena Cano-Fajardo ^{1*} <https://orcid.org/0000-0001-9404-3666>Karla Neidy Perales-Hinojosa ² <https://orcid.org/0009-0001-4581-0184>Mayra Alejandra Mireles-Alonso ³ <https://orcid.org/0000-0002-5277-4476>San Juana López-Guevara ⁴ <https://orcid.org/0000-0002-2984-3576>Jesús Alejandro Guerra-Ordoñez ⁵ <https://orcid.org/0000-0002-4587-3214>Tirso Duran-Badillo ⁶ <https://orcid.org/0000-0002-7274-3511>

1. Doctor in Teaching Methodology, Universidad Autónoma de Tamaulipas, Ciudad Victoria, Tamaulipas, Mexico.

2. Bachelor's Degree in Nursing, Universidad Autónoma de Tamaulipas, Ciudad Victoria, Tamaulipas, Mexico.

3. Master's Degree in Nursing, Universidad Autónoma de Tamaulipas, Ciudad Victoria, Tamaulipas, Mexico.

4. Doctor in Teaching Methodology, Universidad Autónoma de Tamaulipas, Ciudad Victoria, Tamaulipas, Mexico.

5. Doctor in Nursing Science, Universidad Autónoma de Tamaulipas, Ciudad Victoria, Tamaulipas, Mexico.

6. Doctor in Teaching Methodology, Universidad Autónoma de Tamaulipas, Ciudad Victoria, Tamaulipas, Mexico.

*Author for correspondence: lcano@docentes.uat.edu.mx

Received: 25/03/2025**Accepted: 19/10/2025**

Abstract

Introduction: Eating habits are influenced by family, with mothers being primary caregivers and therefore having a decisive influence on the food their school-age children eat. **Objective:** To investigate the relationship between maternal perceptions and eating behaviors and how these influence the nutritional status of schoolchildren. **Methodology:** This descriptive correlational study was conducted with 223 pairs of mothers and students enrolled in grades first to sixth at a public elementary school. Students with motor disabilities, illiterate mothers, or mothers who were not the primary caregiver were excluded. Eating behaviors were measured using the Physical Appearance and Health Questionnaire, which includes maternal perception of body images through pictograms, and the Children's Eating Behavior Questionnaire. Respect for confidentiality, anonymity, assent and informed consent was maintained throughout. **Results:** Only 47 % of mothers correctly identified their children's nutritional status. The remainder underestimated the prevalence of overweight and obesity, with only 34.5 % of schoolchildren being of normal weight. A statistically significant difference was found between maternal perception of schoolchildren's weight and their actual nutritional status $p=0.000$. Additionally, a Spearman correlation was performed, yielding values $p>0.01$. **Conclusions:** Mothers' tendency to underestimate their children's weight can hinder the adoption of healthy habits and increase their risk of developing chronic diseases. Therefore, it is essential to implement educational strategies aimed at mothers to help them identify their children's nutritional status and promote healthy eating practices.

Key words: Maternal perception; Nutritional status; Eating behaviors (DeCS).

Resumen

Introducción: La adopción de hábitos alimentarios se encuentra influenciada por la familia, siendo la madre su cuidador principal por lo que su percepción tiene una influencia decisiva en la clase de alimentos ingeridos por los estudiantes en edad escolar. **Objetivo:** Conocer la relación entre la percepción materna y las conductas alimentarias sobre el estado nutricional del escolar. **Metodología:** Estudio descriptivo correlacional, en 223 diadas de madres y alumnos de primero a sexto año matriculados en institución pública, seleccionadas por muestreo no probabilístico, se excluyeron alumnos con discapacidad motriz, madres analfabetas o que no fuese la cuidadora principal. Se empleó Cuestionario de Apariencia Física y Salud, incluye percepción materna con pictograma de imágenes corporales y el Children's Eating Behaviour Questionnaire para medir conductas alimentarias, se respetó confidencialidad, anonimato, asentimiento y consentimiento informado. **Resultados:** Solo 47 % de madres identificaron correctamente la condición nutricional de sus hijos, el resto subestimó el sobrepeso y la obesidad, solo 34.5 % de los escolares tenía un peso normal. Se encontró una diferencia estadística significativa con $p=0.000$ entre percepción materna sobre el peso del escolar, y el estado nutricional real del mismo. Además, se realizó correlación de Spearman obteniendo valores de $p>0.01$. **Conclusiones:** La subestimación materna del sobrepeso y la obesidad puede dificultar la adopción de hábitos saludables, acrecentando el riesgo de enfermedades crónicas en la infancia. Es esencial implementar estrategias educativas dirigidas a las madres para mejorar la identificación del estado nutricional de sus hijos y fomentar prácticas alimentarias saludables.

Palabras clave: Percepción materna; Estado nutricional; Conductas alimentarias (DeCS).



Abstrato

Introdução: A adoção de hábitos alimentares é influenciada pela família, sendo a mãe a principal cuidadora. Portanto, sua percepção tem influência decisiva sobre os tipos de alimentos consumidos por crianças em idade escolar. **Objetivo:** Determinar a relação entre a percepção materna e os comportamentos alimentares em relação ao estado nutricional de crianças em idade escolar. **Metodologia:** Foi realizado um estudo descritivo e correlacional. Utilizou-se um método de amostragem não probabilística para selecionar 223 díades de mães e alunos do primeiro ao sexto ano matriculados em instituições públicas, selecionadas por amostragem não probabilística. Foram excluídas crianças com deficiência motora, mães analfabetas ou mães que não eram as principais cuidadoras. O Questionário de Aparência Física e Saúde, que inclui a percepção materna com pictogramas de imagem corporal, e o Questionário de Comportamento Alimentar Infantil foram utilizados para avaliar os comportamentos alimentares; a confidencialidade, o anonimato, o assentimento e o consentimento informado foram respeitados. **Resultados:** Apenas 47 % das mães identificaram corretamente o estado nutricional de seus filhos; as demais subestimaram o sobrepeso e a obesidade. Apenas 34,5 % das crianças em idade escolar apresentavam peso normal. Foi encontrada uma diferença significativa de $p = 0,000$ entre a percepção das mães sobre o peso de seus filhos e o estado nutricional real destes. Além disso, foi realizado um teste de correlação de Spearman, que apresentou valores $p > 0,01$. **Conclusões:** A subestimação materna do sobrepeso e da obesidade pode dificultar a adoção de hábitos saudáveis, aumentando o risco de doenças crônicas na infância. Conclui-se que é essencial implementar estratégias educativas direcionadas às mães para melhorar sua capacidade de identificar o estado nutricional de seus filhos e promover práticas alimentares saudáveis.

Palavras-chave: Percepção materna; estado nutricional; comportamentos alimentares (DeCS).

Introduction

Globally, there has been an increase in food-related problems, leading to nutritional imbalances. It is estimated that over 900 million people suffer from malnutrition, with 170 million schoolchildren being underweight ⁽¹⁾. In Mexico, one in three children aged six to eleven is overweight or obese, while 35 % and 2 %, respectively, are very thin ⁽²⁾. In 2016, 33.2 % of schoolchildren were overweight or obese. The nutritional transition in Mexico has led to a change in consumption habits, with healthy products being replaced by energy-dense foods that are high in sugars and fats ⁽³⁾. The high consumption of sugary drinks and low levels of physical activity in schools, coupled with the widespread availability of these foods, are the cause of high rates of overweight and obesity. For example, it is believed that 92 % of children aged 5 and 11 regularly consume sugary



drinks, and 50 % include ultra-processed snacks, sweets and cereals in their diet ⁽³⁾. Consequently, Mexico is considered a country with a malnutrition problem, as it currently has the highest prevalence of childhood obesity in the world ⁽⁴⁾. Eating a healthy diet helps to prevent malnutrition. A healthy diet should be varied and balanced, and tailored to individual characteristics such as age, gender, lifestyle, and level of physical activity. Cultural environment, food availability and local eating patterns can also significantly influence eating behavior. This set of factors plays a fundamental role in understanding malnutrition because it reflects people's relationship with food. This relationship is influenced by direct experiences with food, the family environment, social status, emotional ties and cultural traditions. This has led to a growing interest in analyzing the relationship between obesity and the eating habits of schoolchildren. Changes in the current socioeconomic context have caused both parents and children to adopt unhealthy eating habits, increasing their consumption of processed and fast food, which negatively impacts children's eating habits and nutritional status ⁽⁵⁾.

In addition, low levels of physical activity, coupled with spending long hours watching television and playing video games, are factors that contribute to obesity in schoolchildren ⁽⁶⁾. Nutritional status refers to the body's condition in relation to nutrient intake, which is why diet and nutrition are essential at any stage of life, particularly during the early stages of growth and physiological development ⁽⁷⁾.

The development of eating habits is influenced by three main factors: Family, the media and school. Family is the first environment in which children encounter eating practices, and it exerts a significant influence on their eating-related behavior. These habits are the result of an implicitly shared social and cultural construct among family members ⁽⁶⁾.

Beliefs and dietary practices play a significant role in eating behaviors as they primarily influence mothers' inaccurate perceptions of their children's nutritional status. This, in turn, affects their



children's eating behaviors and self-perception ^(8,9). Studies have examined mothers' inaccurate perceptions of their children's nutritional status (weight/height) and revealed that inadequate maternal perception is a risk factor for the development and maintenance of overweight and obesity. Indeed, 98 % of mothers fail to recognize their children's actual nutritional status ⁽¹⁰⁾.

Some researchers have emphasized the fundamental role of maternal perception ⁽¹¹⁾, given that, historically, a fat child has been considered healthy, while a thin child has been considered unwell ⁽¹²⁾. In Chile, 60 % of mothers of schoolchildren with excessive malnutrition perceived their children as thinner ⁽¹⁰⁾. Similarly, the majority of mothers of overweight or obese children in the Mexican state of Nuevo León underestimated their children's weight ⁽¹¹⁾. As nursing personnel, we are involved in this issue due to the notable increase in the prevalence of childhood overweight and obesity. These conditions could be prevented and modified from an early age by offering education based on the determinants related to this health problem, as well as through visits to child health departments. It is therefore essential to develop strategies to reduce the incidence of these conditions and to promote optimal nutritional status in the school population. This can be achieved by implementing programs that encourage healthy eating habits and contribute to the formation of healthy lifestyles in this group ^(13,14).

Based on the above and following a thorough analysis of the available literature, it is deemed appropriate to conduct further research into these phenomena. The research question is therefore: What is the perception of mothers regarding their children's nutritional status and eating behaviors? with the aim of understanding the relationship between maternal perception and eating behaviors in relation to the nutritional status of schoolchildren.

Methodology

This is a descriptive correlational study. The population consisted of 788 mother-child pairs of school age from first to sixth grade, (aged 6 to 12) from a public elementary school located in the



city of Matamoros, Tamaulipas, from morning and afternoon shifts. The sample size was calculated using a finite population formula with the following parameters: A margin of error of 5 % (0.05), a confidence level of 95 % (1.96) and a standard deviation of 0.5. This yielded a sample of 223 mother-child pairs, obtained by stratified random probability sampling. Each stratum corresponded to a grade level in order to represent the situation at each age and grade level.

The inclusion criteria were: Being enrolled as a student between august 2023 and july 2024; being of either gender; and agreeing to participate. Students, who were not aged between 6 and 12, had a disability that limited their movement, had mothers who were not their primary caregivers, had illiterate mothers, or did not agree to participate in the study were excluded. Finally, cases in which the survey was not completed or the informed consent form was not signed were excluded. A questionnaire was used to collect the following data and information: a) Data on the mother: age, level of education, approximate weight and height, and whether she works? If so, please indicate how many hours, b) Data on the student: Age, sex, grade and group. Subsequently, the Physical Appearance and Health Questionnaire, consisting of 19 items divided into four sections, was administered. For the purposes of this research, only one item from section one, “perception by words”, was considered. Here, the mother could select from very underweight, underweight, normal weight, overweight, or very overweight, to indicate her perception of her child's weight. There was also a section that included a pictogram with seven body images divided into four categories according to the age and sex ranges of the schoolchildren. Here, the mother could select the image that most closely matched her child's body shape ⁽¹⁵⁾. Regarding perception by images, a positive result was considered when mothers of underweight and normal weight children, according to the body mass index (BMI), selected an image representing a size below or above average. Conversely, negative results were categorized as underestimation or overestimation of perception. This questionnaire was validated for the Mexican population in 2009, with a sensitivity



of 9 % and a specificity of 21 % for the perception by words test, and a sensitivity of 76 % and a specificity of 15 % for the perception by images test ⁽¹⁶⁾.

The Children's Eating Behavior Questionnaire (CEBQ) (17), adapted for the Mexican population, provides information on eating behaviors. Consisting of 34 items, it comprises six scales: emotional eating and binge eating (11 items); enjoyment of food (5 items); desire to drink (3 items); moderate appetite (8 items); emotional fasting (4 items); and pickiness (3 items). The questionnaire has a Cronbach's alpha reliability of 0.882 (18), whereas in this study it was 0.822. Finally, a SECA scale, model 813, with a capacity of 150 kg and an accuracy of 0.1 kg, was used to determine weight. BMI was calculated using the table in the National Health Card (Cartilla Nacional de Salud) for children aged zero to nine and ten to nineteen. Height was measured using a SECA 213 stadiometer with a metric scale.

This study was approved by the Research and Ethics Committee of the Matamoros Multidisciplinary Academic Unit under the reference number 119. Authorization was then requested from the educational institution's management to carry out data collection. The study was conducted in accordance with the General Health Law Regulations, particularly the section on research involving human subjects ⁽¹⁹⁾, which stipulates that all health research must take into account ethical considerations such as dignity, consent, confidentiality, anonymity and informed consent.

The statistical package Statistical Package for the Social Sciences (SPSS) version 24 for Windows was used for data analysis. Descriptive statistics were used in the form of frequencies and percentages, as well as Spearman's nonparametric correlation test, in order to determine the relationship between the variables.



Results

The BMI of the mother and child were 26 % and 34.5 %, respectively. These were both within the normal weight range, (Table 1).

Table 1. Nutritional status of mothers and schoolchildren, 2024 (n=223)

Variable	<i>f</i>	%
BMI-Mother		
Underweight	1	0.4
Normal Weight	58	26.0
Overweight	103	46.2
Obesity I	42	18.8
Obesity II	14	6.3
Obesity III	5	2.2
BMI-Schoolchildren		
Underweight	59	26.5
Normal Weight	77	34.5
Overweight	42	18.8
Obesity	45	20.2

Source: Self-development

Table 2, shows that only 47 % of mothers correctly perceived their child's nutritional status. It is particularly notable that mothers tended to misperceive children of normal weight as underweight or overweight, and children who were overweight or obese as normal weight or underweight.

Table 2. Choice of body images and body mass index (BMI), 2024 (n=223)

Choice of body images	Overweight u Obesity		Normal Weight		Under Weight	
IMC	<i>f</i>	%	<i>f</i>	%	<i>f</i>	%
Underweight	1	0.4	2	0.8	8	3.6
Normal Weight	5	2.2	59	26.4	62	27.8
Overweight	11	4.0	23	10.3	7	3.1
Obesity	27	12.1	16	7.1	2	0.8

Source: Self-development

To address the overall objective, a Kruskal–Wallis non-parametric test was performed to determine whether there was a significant difference between the BMI of the schoolchildren and their mothers' choices of body images. The result obtained was $p=0.000$, establishing a statistically significant difference between mothers' perceptions of their children's weight and their actual nutritional status, (Table 3).



Table 3. Difference between BMI and images, 2024

Kruskal-Wallis Test		
X ²	df	p
92.958	2	.000

Source: Self-development. Notes: X² = Chi-square; gL= grades of freedom; p= significance

Finally, to determine whether there was a relationship between eating behaviors and the nutritional status of the schoolchildren, a Spearman's nonparametric test was performed. Values greater than $p= 0.01$ were obtained, which indicates that there is no statistical relationship between the two variables. Therefore, regardless of their classification, one does not affect the other, (Table 4).

Table 4. Spearman's nonparametric test, 2024

Variable	Pro-ingestion		Anti-ingestion		BMI	
	p	Significance	p	Significance	p	Significance
Pro-ingestion			.271**	.000	.031	.641
Anti-ingestion	.271**	.000			-0.85	.208
BMI	0.31	.641	-0.85	.208		

Source: Self-development

Discussion

It is important to note that most of the mothers in this study had an imbalanced nutritional status, primarily in the form of overweight and obesity. These figures are consistent with the results of the 2020-2023 National Health and Nutrition Survey (Encuesta Nacional de Salud y Nutrición ENSANUT) ⁽²⁰⁾, this influences their distorted perception of their children's weight, which could be considered a limitation, indeed, just over half of mothers incorrectly estimated the weight of the schoolchild, underestimating it.

It should be noted that the results of this study only reflect the perceptions of the surveyed couples. Nevertheless, these findings have practical implications for the nursing profession, as they emphasize the importance of establishing healthy habits. These data are consistent with previous research. In studies conducted in Shanghai, China and Brazil ^(9,22), mothers whose weight was outside the normal range tended to underestimate their children's weight, this phenomenon was



observed in nearly a quarter of the interviewed mothers. Having an inaccurate perception of their children's weight can lead parents to encourage unhealthy eating habits, which disrupts with family eating dynamics and can cause weight disorders, limit physical activity and promote a sedentary lifestyle.

Obesity is a global public health problem caused by the nutrient-deficient diets that mothers provide for their children, the environment, in which schoolchildren live, including their family, school and friends, also influences their food choices, furthermore, sedentary modern lifestyles do not encourage healthy habits among schoolchildren. Only one-third of schoolchildren undergoing somatometry were found to be of normal weight, this is not an isolated finding, as various studies conducted in different regions of the world have revealed the same pattern ^(5,6,10,23,24).

This study shows that nearly half of the mothers accurately assessed their children's nutritional status, tending to underestimate cases of overweight and obesity. This phenomenon has been observed in previous research and may be due to cultural factors or the normalization of being overweight. If childhood overweight and obesity are underestimated, it could hinder the implementation of changes to family habits that are needed to prevent and control this disease. This is consistent with previous studies which showed that most overweight or obese mothers have an inaccurate perception of their children's weight and that this distortion is a consequence of the steady increase in obesity among schoolchildren ^(10,11). A study conducted in Romania assessed schoolchildren's weight perception from the perspective of their guardians and found that more than half misperceived their child's actual weight, where three-fifths of the schoolchildren were actually at a normal weight ⁽²⁵⁾. Furthermore, a study conducted in Argentina found that mothers' perceptions of their obese children are distorted ⁽²⁶⁾, although schoolchildren of this age are independent, it is the responsibility of parents to ensure their children are well-nourished, which is why it is important for them to have a realistic perception of their children's weight.



To determine whether eating behaviors are related to the nutritional status of schoolchildren, the six dimensions of the child eating behavior questionnaire were analyzed, no statistically significant relationship was found. This suggests that eating behaviors are not directly related to BMI in this group of children.

Food is one of the most fundamental human needs and influences emotional, cognitive and behavioral aspects related to food intake, including needs, pleasure, sociocultural patterns and the social environment, all of these factors affect food intake, which can increase or decrease, causing problems with weight control and leading to inappropriate eating behavior ^(27,28).

The results reveal the behavior of mothers and children in an educational institution in the city of Matamoros, however, it cannot be guaranteed that the findings represent the behavior of the wider population, therefore, further research into eating behavior issues in schoolchildren is important, this would provide the necessary information to design food education strategies for mothers and schoolchildren.

Conclusions

The findings of this study highlight a significant discrepancy between maternal perception and the actual nutritional status of participating schoolchildren, a high percentage of mothers tended to underestimate their children's weight, particularly in cases of overweight and obesity, this finding is consistent with previous research and suggests a possible cultural normalization of excessive weight within the family environment. This underestimation poses an obstacle to implementing healthy eating strategies and adopting an active lifestyle at home. Misperceptions can hinder the recognition of the need for dietary changes and physical activity among schoolchildren; they can also create barriers to carrying out nurses' recommendations to mothers.

Life expectancy today is a success thanks to interventions related to healthy habits, science, and technology. For its part, the nursing profession guides humanity through its different stages of life,



reducing exposure to risk conditions and thus ensuring a better quality of life, however, despite the efforts of healthcare personnel, current statistical data are not favorable, and trends do not offer an encouraging picture.

It is important to highlight that one of nursing's many roles in society is to promote health, through this, individuals, families and communities are educated in strengthening self-care, this involves a series of actions that address all of a person's needs holistically.

Addressing maternal perceptions of children's physical condition through questions and images places the mother in an analysis of how she perceives her child and the ideal. It should be noted that there are multiple factors that influence this perception, which is why it is suggested that future studies not only address the dyad, but the entire family unit and circles of friends.

Conflict of interest

The authors stated that they have no conflict of interest.

Funding

The lead author was awarded a postgraduate scholarship by the National Council for Humanities, Sciences and Technologies (Consejo Nacional de Humanidades, Ciencias y Tecnologías).

Artificial intelligence

The authors stated that they have not used any artificial intelligence resources in any of the sections of this document.

References bibliographic

1. World Health Organization (WHO). Obesity and Overweight. Informative note [Internet]. OMS;2025 [cited 15 jul 2025]. Available at: <https://www.who.int/es/news-room/fact-sheets/detail/obesity-and-overweight>
2. Fondo de las Naciones Unidas para la infancia (UNICEF). El índice mundial de obesidad supera por primera vez al de bajo peso entre los niños y niñas en edad escolar y adolescentes (Siglas). [Internet]. 2025 [cited 30 sep 2025]. Available at: <https://goo.su/ofiWD>
3. Gobierno de México. Obesidad infantil: nuestra nueva pandemia [Internet]. Hablemos de salud 2021 [cited 21 mar 2025]. Available at: <https://www.gob.mx/promosalud/articulos/obesidad-infantil-nuestra-nueva-pandemia?idiom=es>



4. Instituto Nacional de Salud Pública (INEGI). Encuesta Nacional de Salud y Nutrición 2018: Presentación de resultados [Internet]. México; 2019 [cited 21 mar 2025]. Available at: https://ensanut.insp.mx/encuestas/ensanut2018/doctos/informes/ensanut_2018_presentacion_resultados.pdf
5. Encina VC, Gallegos LD, Espinoza MP, Arredondo GD, Palacios CK. Comparación de la conducta alimentaria en niños de diferentes establecimientos educacionales y estado nutricional. *Revista chilena de nutrición* [Internet]. 2019 [cited 30 sep 2021];46(3):254-263. Available at: <https://dialnet.unirioja.es/servlet/articulo?codigo=9541644>
6. Sánchez CB, Reyes CZ, Barajas GL, García OL, Valdés ME. Hábitos alimentarios y actividad física en población escolar con normopeso, sobrepeso, obesidad y resistencia a la insulina. *Journal of Behavior and Feeding* [Internet]. 2022 [cited 13 feb 2023];2(2):20-27. Available at: <https://doi.org/10.32870/jbf.v2i2.30>
7. Saintila J, Villacís JE. Estado nutricional antropométrico, nivel socioeconómico y rendimiento académico en niños escolares de 6 a 12 años. *Revista Nutrición Clínica y Dietética Hospitalaria* [Internet]. 2020 [cited 13 feb 2021];40(1):74-81. Available at: <https://revista.nutricion.org/PDF/SAINTILA.pdf>
8. Ayine P, Selvaraju V, Venkatapoorana CM, Bao Y, Gaillard P, Geetha T. Eating behaviors in relation to child weight status and maternal education. *Children* [Internet]. 2021 [cited 23 sep 2025];8(32):1-9. Available at: <https://pubmed.ncbi.nlm.nih.gov/33430408/>
9. Mattos MH, Pandolfi MM, Nunes FC, Fonseca SC, De Meneses GG, Cavalcante TG, et al. Percepção materna do estado nutricional do filho: Um caminho para o diagnóstico precoce da obesidade infantil. *RBONE* [Internet]. 2022 [cited 23 mar 2025];15(95):593-601. Available at: <https://www.rbone.com.br/index.php/rbone/article/view/1472/1137>
10. Rozas K, Huerta P, Planett J, Arancibia M, Araya MV. Alteración de la percepción materna sobre el estado nutricional de sus hijos ¿nuevo factor de riesgo cardiovascular? *Rev Chil Cardiol* [Internet]. 2020 [cited 21 mar 2025];39(3):216-222. Available at: <http://dx.doi.org/10.4067/S0718-85602020000300216>
11. Flores-Peña Y, Avila-Alpírez H. Percepción materna del peso del hijo, problemas del estilo de vida y autoeficacia para manejarlos. *Aquichan* [Internet]. 2021 [cited 18 mar 2025];21(2):1-12. Available at: <https://doi.org/10.5294/aqui.2021.21.2.8>
12. Wang J, Zhu D, Cheng X, Liuzhou Y, Zhu B, Montgomery S, et al. Maternal perception of child weight and concern about child overweight mediates the relationship between child weight and feeding practices. *Public Health Nutr* [Internet]. 2022 [cited 23 sep 2025];25(7):1780-1789. Available at: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9991628/>
13. Segura-Álvarez M, Restrepo-Segura Y, Ramírez-Valencia M. Sobrepeso y obesidad: una mirada desde la educación para la salud, en un sistema de bienestar universitario. *Rev. Salud Pública* [Internet]. 2024 [cited 26 sep 2025];26(6):1-7. Available at: <https://doi.org/10.15446/rsap.v26n6.116166>
14. Mosquera HM, Mosquera HM, De Armas DL, Brito RY. Estado nutricional y hábitos alimenticios en niños de un colegio público de Valledupar. *Rev. Méd Risaralda* [Internet]. 2016 [cited 18 oct 2021];22(1):42-48. Available at: http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S0122-06672016000100007
14. Ekstein KC, Mikhail LM, Ariza AJ, Scott TJ, Millard SC, Binns HJ. Parents' perceptions of their child's weight and health. *American Academy of Pediatrics* [Internet]. 2006 [cited 16 mar 2025];117(3):681-690. Available at: <https://doi.org/10.1542/peds.2005-0910>



15. Flores-Peña Y, Trejo-Ortiz P, Gallegos-Cabriales E, Cerda-Flores R. Validez de dos pruebas para evaluar la percepción materna del peso del hijo. *Salud Pública México* [Internet]. 2009 [cited 21 mar 2025];51(6):489-495. Available at: https://www.scielo.org.mx/scielo.php?script=sci_arttext&pid=S0036-3634200900060000
16. Wardle J, Guthrie CA, Sanderson S, Rapoport L. Development of the children's eating behaviour questionnaire. *Journal of Child Psychology Psychiatry* [Internet]. 2001 [cited 13 feb 2025]; 42(7):963-970. Available at: <https://doi.org/10.1111/1469-7610.00792>
17. Vázquez-Pérez L, González-Herrera L, Hattori-Hara M, Zavala-Cervantes A. Traducción y validación del children's eating behavior questionnaire (CEBQ). *Ciencia y humanismo en la salud* [Internet]. 2020 [cited 10 apr 2025];7(1):1-9. Available at: <https://revistacienciayhumanismoenlasalud.uady.mx/cienciayhumanismo/article/view/136>
18. Diario Oficial de la Federación. Proyecto de norma oficial mexicana PROY-NOM-012-SSA3-2007, Que establece los criterios para la ejecución de proyectos de investigación para la salud en seres humano. Secretaría de Salud [Internet]. México; 2007 [cited 07 dec 2024]. Available at: <http://www.dof.gob.mx/normasOficiales/3880/salud/salud.htm>
19. Secretaría de Salud. Encuesta Nacional de Salud y Nutrición. Resultados de la ENSANUT 2020-2023. Instituto Nacional de Salud Pública [Internet]. Gobierno de México; 2024 [cited 23 mar 2025]. Available at: <https://www.insp.mx/avisos/presentan-resultados-de-la-ensanut-2020-2023>
20. Centro de Investigación en Evaluación y Encuestas (CIEE). Encuesta Nacional de Salud y Nutrición. Encuestas [Internet]. México; 2023. [cited 23 mar 2025]. Available at: <https://ensanut.insp.mx/encuestas/ensanutcontinua2023/index.php>
21. Wang J, Zhu D, Cheng X, Liuzhou Y, Zhu B, Montgomery S, et al. Maternal perception of child weight and concern about child overweight mediates the relationship between child weight and feeding practices. *Public Health Nutrition* [Internet]. 2022 [cited 15 may 2025];25(7):1780-1789. Available at: <https://pubmed.ncbi.nlm.nih.gov/35000661/>
22. Terry BB, Rodríguez VL, Silvera TD, Rodríguez FV, Chávez VHN, Rodríguez SA. Sobrepeso, obesidad y conductas alimentarias en escolares de primaria, municipio Cruces, Cienfuegos. *Revista Cubana Salud Pública* [Internet]. 2021 [cited 18 mar 2025];47(1):1-21. Available at: http://scielo.sld.cu/scielo.php?script=sci_arttext&pid=S0864-34662021000100014
23. Morales PA, Rodríguez RI, Massip NJ, Sardinias AM, Balado SR, Morales VL. Preferencias y conductas alimentarias en adolescentes de secundaria básica. *Revista Cubana Pediatría* [Internet]. 2021 [cited 23 mar 2025];93(2):1-15. Available at: http://scielo.sld.cu/scielo.php?script=sci_arttext&pid=S0034-75312021000200005&lng=es
24. Trandafir AV, Fraseniu M, Lotrean LM. Assessment of actual weight, perceived weight and desired weight of Romanian school children-opinions and practices of children and their parents. *Int J Environ Res Public Health* [Internet]. 2022 [cited 20 nov 2024];19(6):1-16. Available at: <https://doi.org/10.3390/ijerph19063502>
25. Muñante RE. Percepción del estado nutricional de los hijos y su relación con el sobrepeso y obesidad infantil en dos colegios de Ica. [Tesis Licenciatura]. Ica Perú: Universidad Nacional San Luis Gonzaga [Internet]. 2023 [cited 30 sep 2025]. Available at: <https://repositorio.unica.edu.pe/items/8551cbd7-5db1-4c7e-b216-286855399a15>
26. Bautista-Díaz M, Castelán-Olivares A, Martín-Tovar A, Franco-Paredes K, Mancilla-Díaz J. Conductas alimentarias de riesgo, percepción de prácticas parentales y conducta asertiva en estudiantes de preparatoria. *Interacciones Revista de avances en psicología* [Internet]. 2020 [cited 18 feb 2022];6(3):1-10. Available at: <https://doi.org/10.24016/2020.v6n3.162>



27. Jimeno-Martínez A, Maneschy I, Rupérez AI, Moreno LA. Factores determinantes del comportamiento alimentario y su impacto sobre la ingesta y la obesidad en niños. JBF [Internet]. 2021 [cited 30 sep 2025];1(1):60-71. Available at: <https://doi.org/10.32870/jbf.v1i1.20>

How to referece this article: Cano-Fajardo L, Perales-Hinojosa K, Mireles-Alonso M, López-Guevara S, Guerra-Ordoñez J, Duran-Badillo T. Relación de la percepción materna y conductas alimentarias sobre el estado nutricional del escolar. SANUS [Internet]. 2025 [cited dd mmm yyyy];10(21):e573. Available at: DOI/URL.

